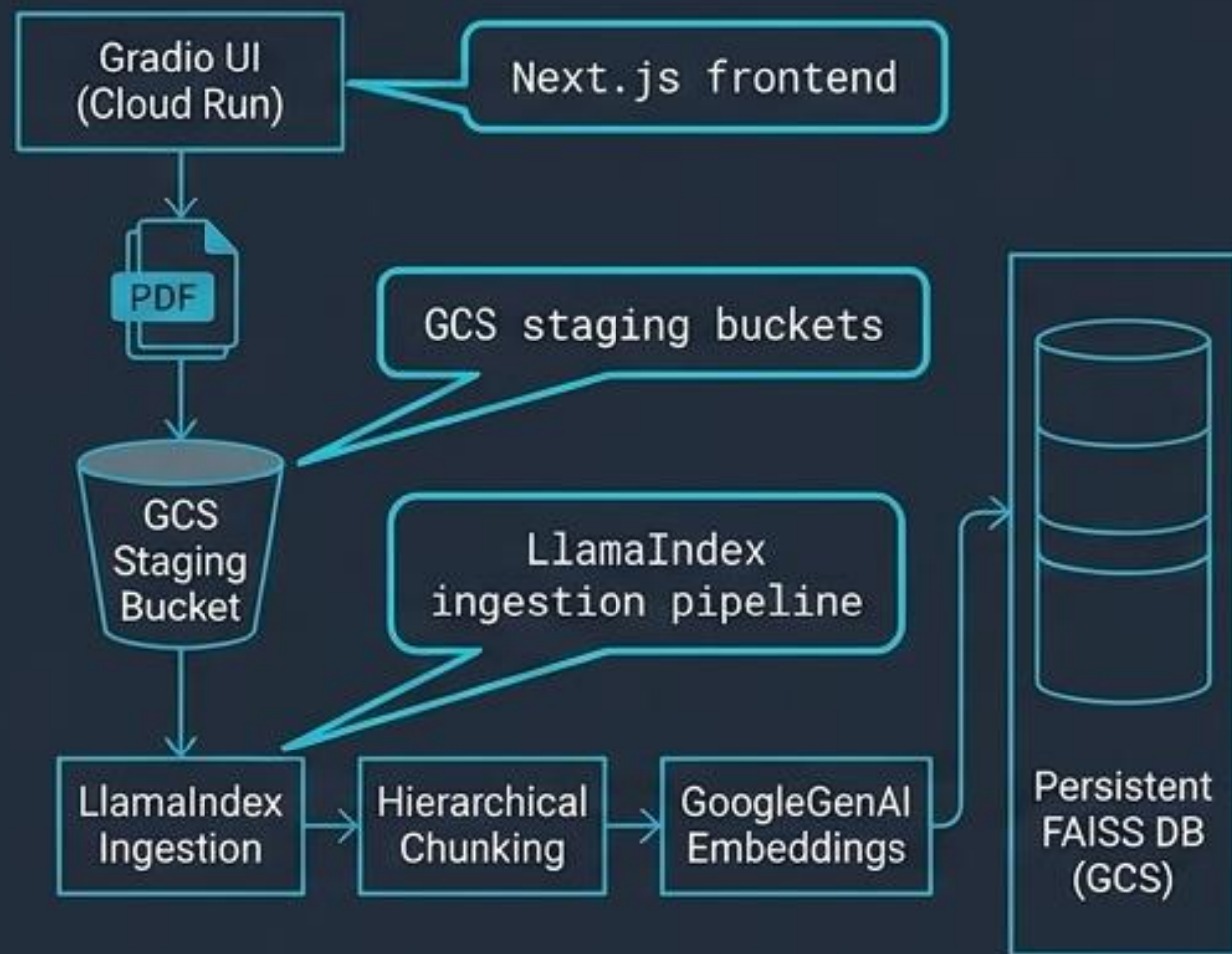


# Enterprise Cloud-Native System Architecture

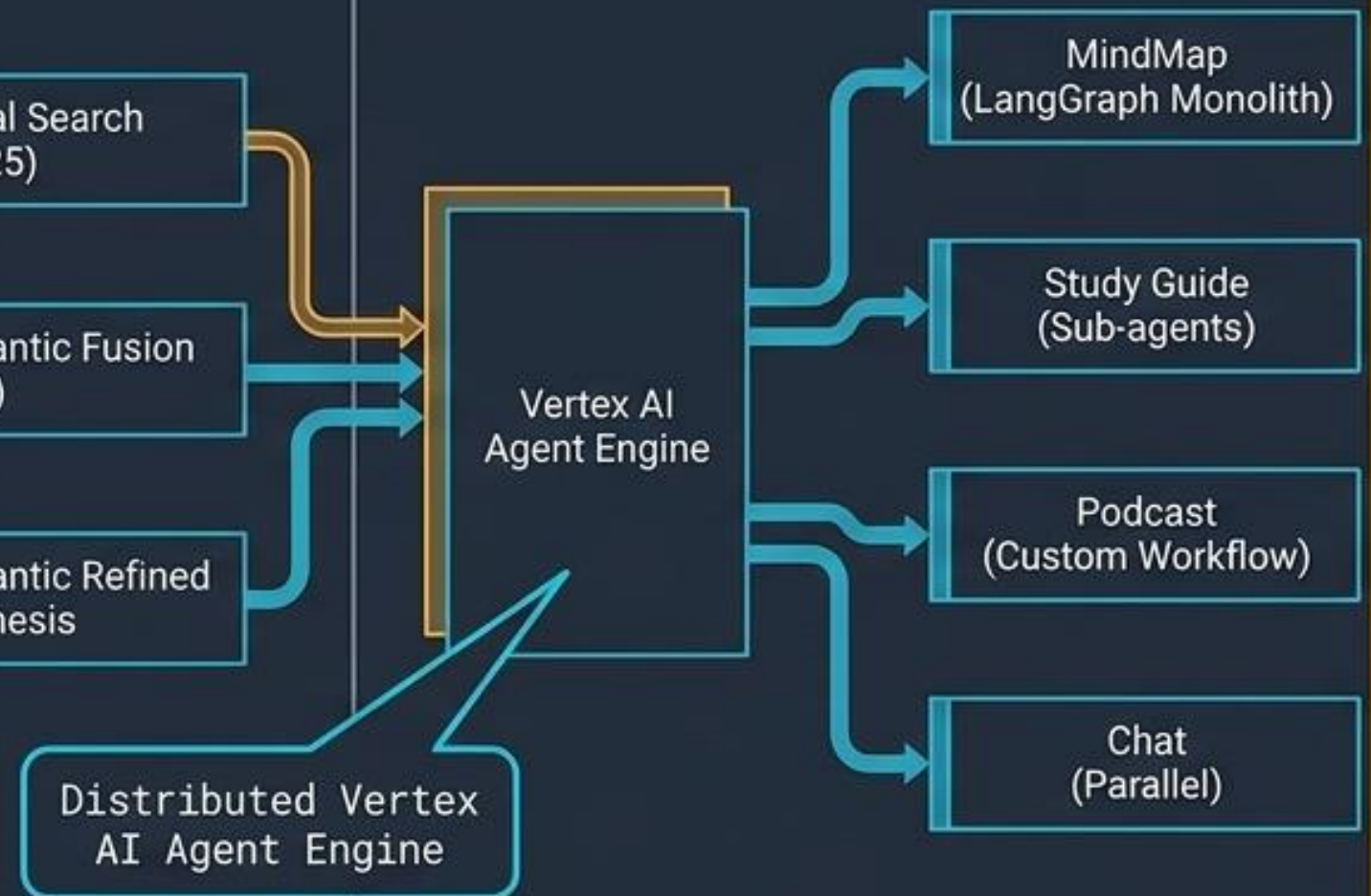
## Ingestion & State



## Retrieval (The 3 Paths)



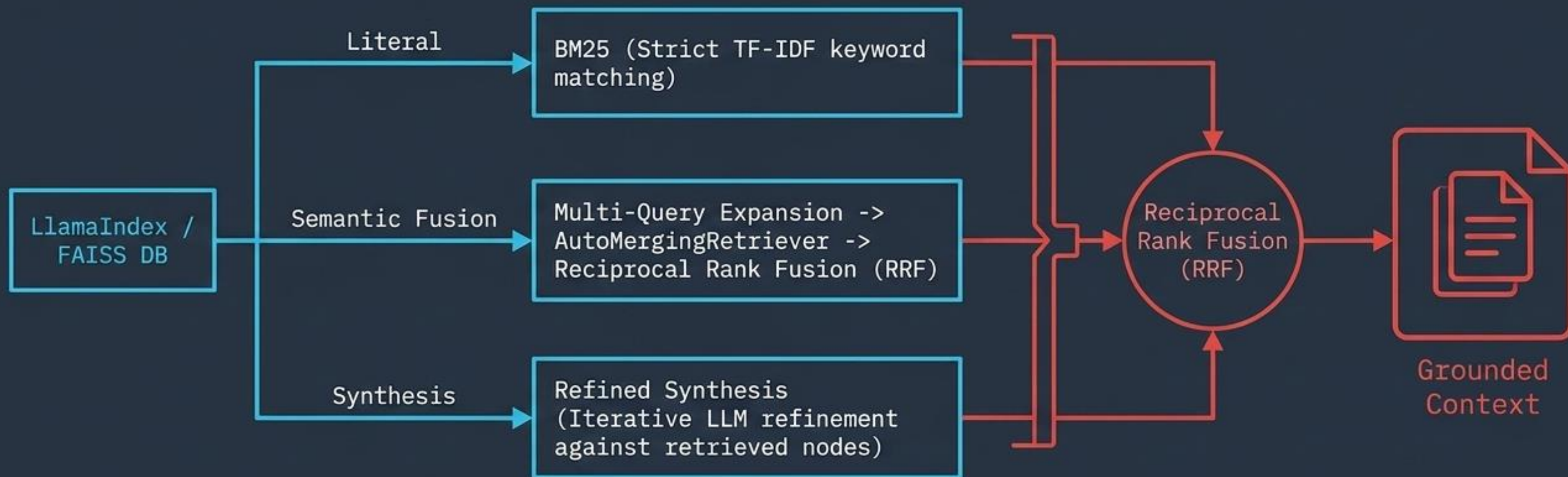
## Intelligence (Vertex AI)



The Initial State: Conceptually pure during local validation, but destined to clash with serverless production physics.



# Eliminating Hallucinations via Tri-Path Search



Dynamic Parallel Querying

Fault-Tolerant Hallucination Elimination

Tech Stack

LlamaIndex

GoogleGenAI Embeddings

FAISS

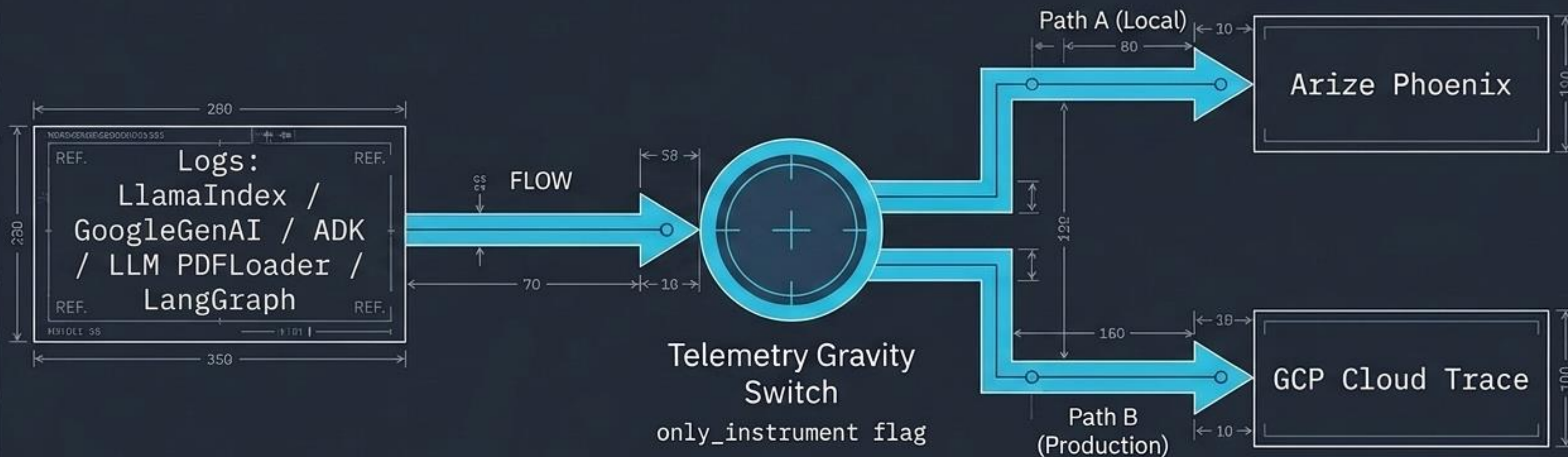
BM25

RRF



# Telemetry Gravity and the Unified Observability Pipeline

A single flag dynamically hijacks the instrumentation pipeline, routing granular spans to local or production environments without altering the execution code.



Tech Stack

OpenTelemetry

Arize Phoenix

GCP Cloud Trace

Python Iterators



# High-Fidelity Custom Quality Assurance Evaluations



## Standard ADK Registry Limitation

Black-Box Failure: Built-in evaluators completely crashed when introspecting custom Google ADK workflows.

Replaced 287 lines of broken framework code with a 10-line Python iterator. Achieved 91% aggregate pass rate.



## Custom JSON Rubric Evaluator

Engineered an "LLM-as-a-Judge" pipeline against strict manual JSON rubrics.

Tech Stack

LLM-as-a-Judge

ADK Evaluators

Custom JSON Rubrics

Gemini 2.5 Pro



# FinOps Governance and the Nuclear Teardown Protocol



Vertex AI Agent Engine continuously billed for "warm" Zombie reasoning containers during downtime.

## The Protocol

- ✓ 1. Scale Cloud Run to zero.
- ✓ 2. Delete Vertex AI reasoning engines via SDK.
- ✓ 3. Purge Artifact Registry builds.
- ✓ 4. Revoke IAM and API access permissions.

Tech Stack

Python SDK

GCP Billing

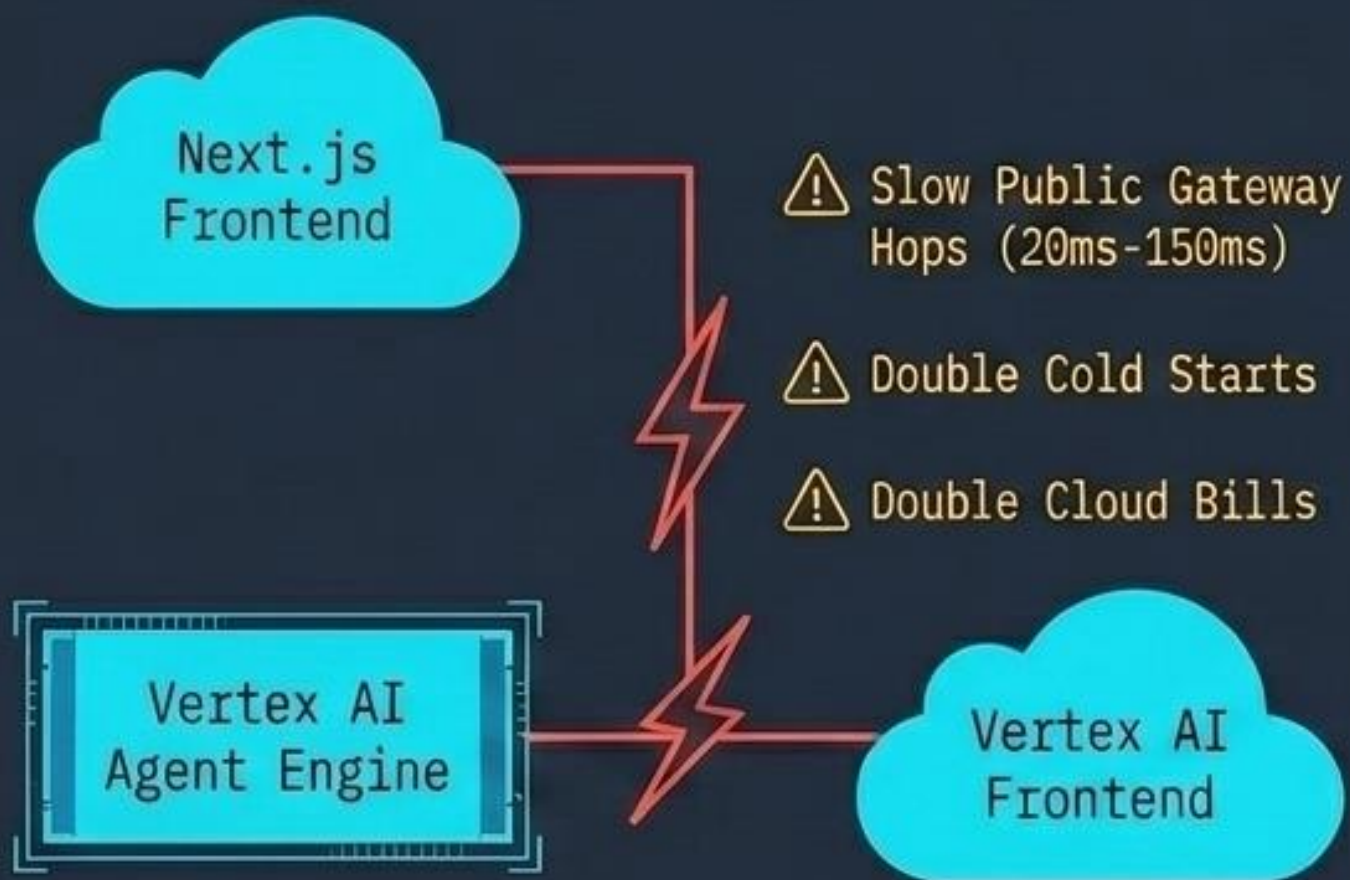
FinOps Automation

IAM Restructuring

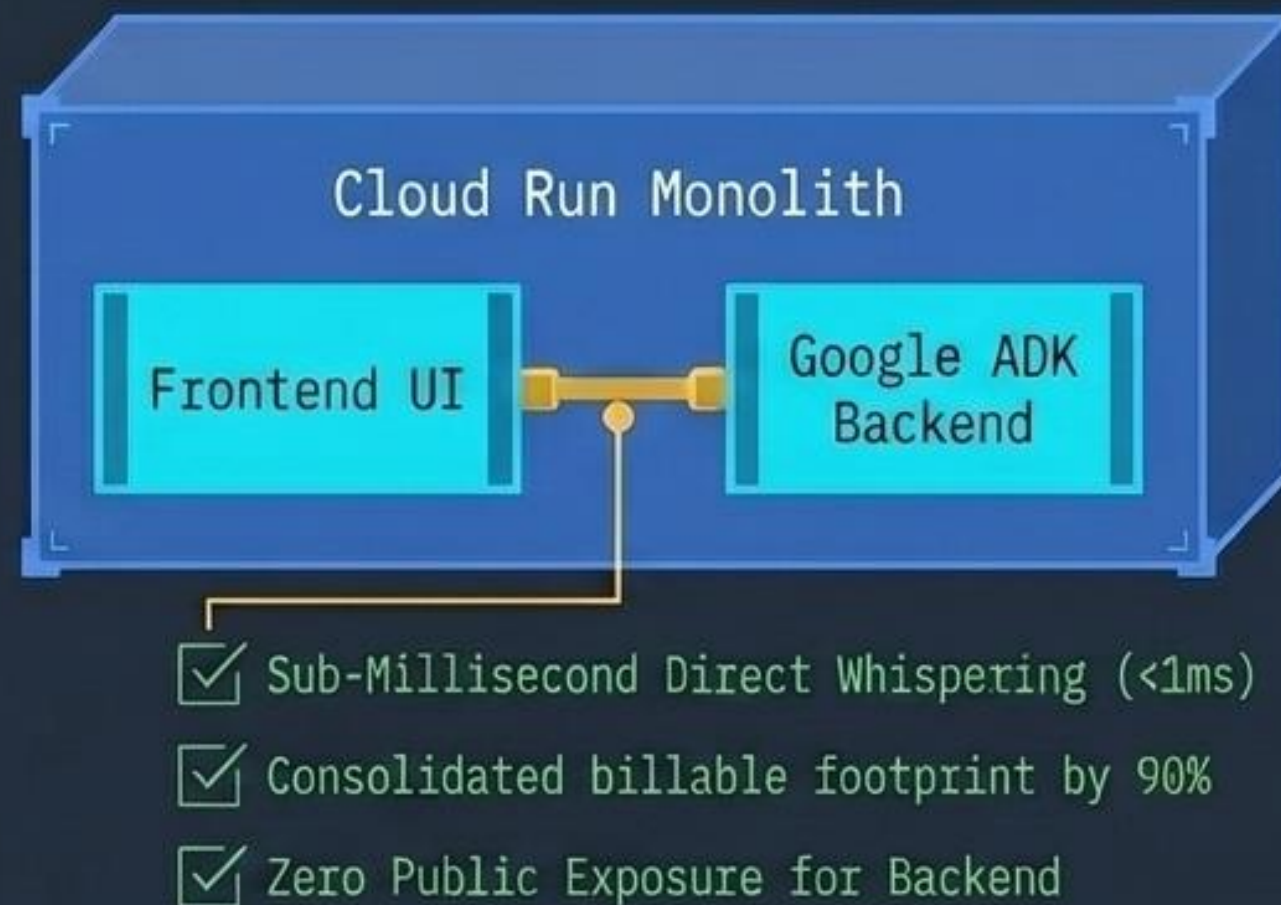


# The Enterprise Pivot: Single-Container Monolith Topology

## Before (Scattered Islands)



## After (Unified Fortress)



Tech Stack

Multi-Stage Docker

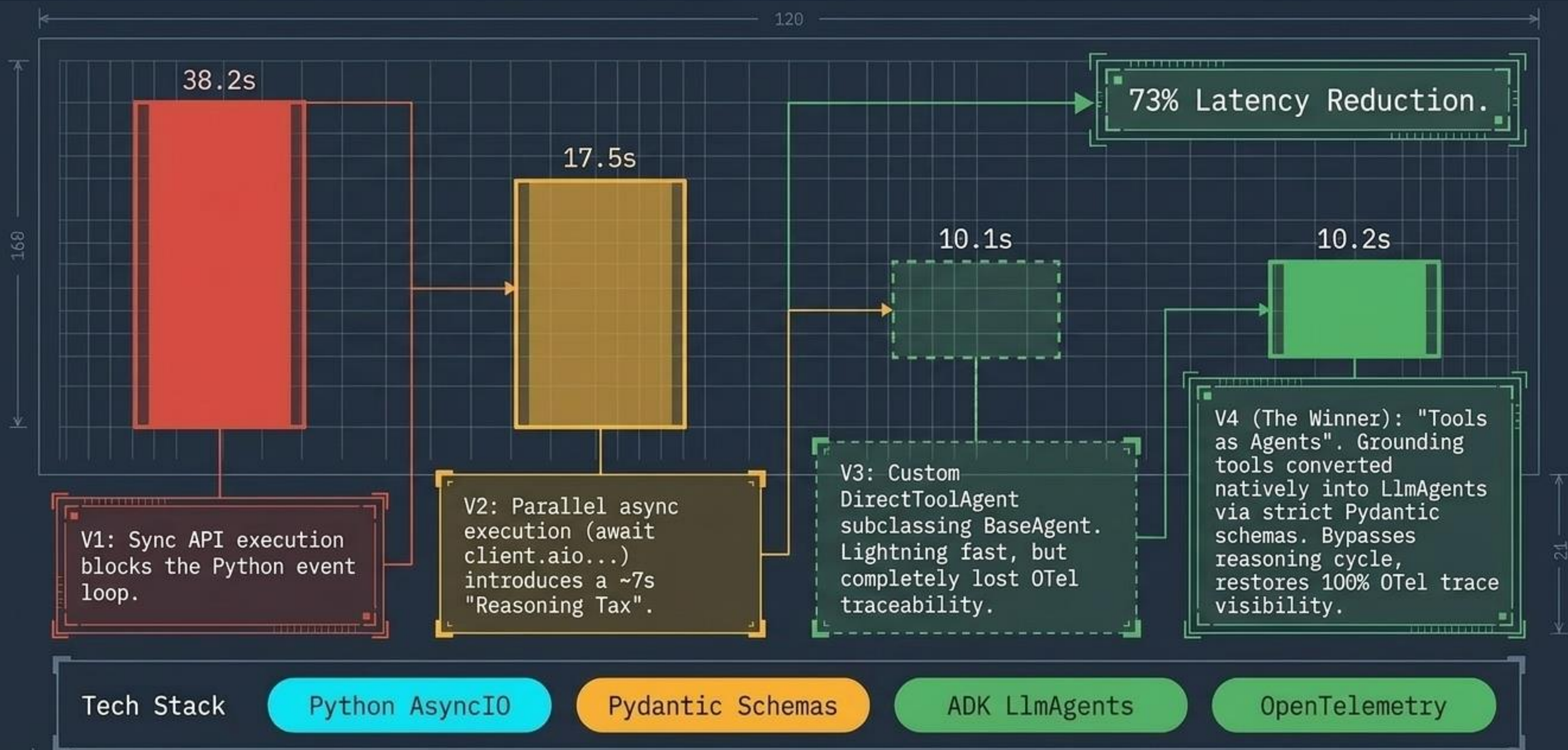
POSIX Process Multiplexer

Private TCP Loopback

Unix Init Traps



# The RAG Latency Optimization Waterfall





# Hybrid Networking: The Cloud-to-Edge Bridge

## 3 Min Generation for 20 Min Audio



### Symptom:

Long-Running Operations (LROs) causing massive 504 Deadline Exceeded timeouts.

### Remedy:

Ejected heavy MLX-Audio synthesis to isolated FastMCP server via self-polling architecture. Tuned polling intervals (2s normal / 8s error) to prevent proxy DDOS crashes.

Tech Stack

FastMCP

ZioK Reserved Tunnel

Server-Sent Events (SSE)

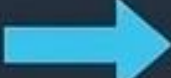
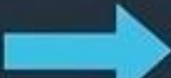
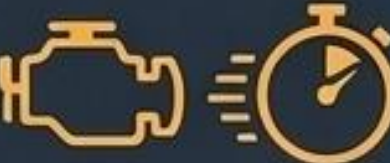
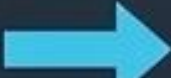
MLX-Audio

Apple Silicon



# Conquering Container Physics and Zero-Trust Security

## Physics Diagnostic Matrix

|                                                                                                                                           |                                                                                     |                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CHALLENGE:</b><br><br>Port Collisions                 |    | <b>SOLUTION:</b><br><br>UNIFIED, ALIGNED NETWORK FLOW<br>Fixed via <b>Ingress Realignment</b> (forcing frontend-to-backend API calls over a sub-millisecond private loopback). |
| <b>CHALLENGE:</b><br><br>Unexpected Process Termination |   | <b>SOLUTION:</b><br><br>SIGNAL SHIELDING<br>Fixed via <b>Signal Shielding</b> (wrapper preventing silent background process death).                                           |
| <b>CHALLENGE:</b><br><br>Slow Service Startup          |  | <b>SOLUTION:</b><br><br>10s PREDICTABLE STARTUP SYNC<br>Fixed via a <b>10s Predictable Startup Sync</b> within the container lifecycle.                                      |

## Security Architecture Vaults



### Tech Stack

Linux POSIX Signals

Port Mapping Alignment

Boot Synchronization

Zero-Trust Local Routing